SCIENCE 1.4 **Name:**_____

WORKSHEET THREE **ACIDS AND ALKALIS**

1. The taste of an acid is _____ The taste of a base is _____(A)
2. When red litmus is put into acids what happens? _____(A)
   When blue litmus is put into acids what happens?_____(A)
   When red litmus is put into bases what happens? _____(A)
   When blue litmus is put into bases what happens? _____(A)
3. Litmus is an example of an acid-base…. _____(A)
4. The name of a vegetable that contains an indicator is … _____(A)
5. The name of a flower that contains an indicator is… _____(A)
6. An indicator is a substance that changes _____ in acids and bases. (A)
7. In what major way is Universal indicator different from Litmus? (A)

   _____

   _____
8. What colour is U.I. in pure water. _____(A)
9. In order to describe how acidic an acid is, or how basic a base is we use the _____ scale. (A)
10. Universal indicator turns different colours depending on the _____ of the solution. (A)
11. Water has a pH of _____ and is said to be… _____(A)
12. If a chemical has a pH of 3 it would be an … _____(A)
13. If a chemical had a pH of 13 it would be a …_____(A)
14. Chemical A has a pH of 9 and chemical B has a pH of 12. A and B are both _____ (acids/bases) but B is a _____ (stronger / weaker) _____ (acid / base) than A. (A,M)
15. Write the names and formulae of the three **strong mineral acids** we use in the laboratory. (M)

    _____

    _____

    _____
16. Write the name and formula of the **weak organic acid** present in vinegar.(M)

    _____
17. What is the definition of a base? (M) _____

    _____
18. What is the definition of an alkali? _____(A)

19. (a) When you put a zinc into acid an invisible gas called _____ is made.(A)

(b) This gas is very _____ (light / heavy). (A)

(c) To test for this gas we put a burning _____ into it and there is a _____ sound. (A)

20. What is the name and formula of a common strong alkali used in the lab. (A)

_____

21. We often refer to sodium hydroxide as **caustic soda**. What is the meaning of **caustic**? (M)

_____

22. A common weak alkali used in the lab is actually a solution made by dissolving a smelly gas in water. What is it called? _____ (A)

23. Two solutions have lost their labels but one of them is vinegar and the other is caustic soda. Explain how you would discover which was which? _____

_____ (M)

24. Some text books suggest that litmus indicator has a **purple** colour in water. How would you account for this?_____ (M)

25. A solution of hydrochloric acid has a pH of 1. A large volume of water is added to the acid and mixed thoroughly. Suggest if there will be any change in the pH of the mixture.

_____(M)

Use this idea to suggest how acid spills should be treated. _____(A)

Suggest another chemical that could be added to reduce the danger of an acid spill.

_____(M)

26. Bases are sometimes classified as the oxides and hydroxides of metals. Use this idea and what you know about acids to help you circle the substances in the list that are bases. (A)

NaCl NaOH K_2O $Zn(OH)_2$ HCl HNO_3 CuO MgSO4 MgO $Fe(OH)_2$

27. Circle the substances in the following list that are acids. (A)

Li_2SO_4 $AlCl_3$ HNO_3 NH_3 Fe_2O_3 H_2SO_4 CH_3COOH NaOH HCl K_2O

28. When acids are reacted with carbonates and bicarbonates a colourless gas is given off.

(a) What is the name of this gas? _____(A)

(b) Describe how you would test for this gas._____(A)

_____

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